

Tal Ben Ami

Performance Engineer | Experienced with LLM Inference & GPU Optimizations

054-9564820 · talba225@gmail.com · TalBenAmi.com · linkedin.com/in/tal-ben-ami · github.com/TalBenAmi

Available 3-4 days/week. Full-time from March 2027 on graduation

Intel performance engineer focused on firmware latency and CPU bottlenecks. Independent projects include an agent harness for llm-d benchmarking on Kubernetes and implementations of CoLA/GaLore research methods for memory-efficient Vision Transformer pre-training.

EXPERIENCE

Intel | Performance Engineer, TDX Design Team

Jan 2026 - Present

- Identified and resolved API pipeline bottlenecks involving MOVDIR64B/non-temporal memory operations and SEAMCALL/SEAMRET transitions. Affected SEAMCALL/TDCALL API latencies on a newer server platform improved from 20-70% higher than Granite Rapids to 20-50% lower (~33-71% overall reduction).
- Owned the team's performance infrastructure and tooling for measuring SEAMCALL and TDCALL API latency. Refactored benchmarks to reduce run-to-run variation and improve measurement reproducibility.
- Built test automation integrating an LLM via the OpenAI API to analyze Git diffs between releases and help identify causes of performance regressions.

SELECTED PROJECTS

llm-d: Inference Benchmarking Agent Harness & MCP Server

Harness · MCP

- Built an agent harness for llm-d, an open-source distributed inference project, translating workload goals into benchmark configurations and orchestrating Kubernetes Jobs and parameter sweeps. Exposed workflows through an MCP server.
- Implemented schema-validated analysis of time to first token (TTFT), inter-token latency (ITL), and throughput, with SLO checks and Pareto comparisons.

CoLA + GaLore: Memory-Efficient Vision Transformers

GitHub

- Developed a PyTorch implementation of CoLA low-rank layers and GaLore gradient projection from research papers. Profiling a large ViT model showed ~70% lower peak allocated GPU memory (>20 to 6.45 GiB) with ~25% higher training time per iteration versus the AdamW baseline.
- On CIFAR-10, a smaller ViT model used ~80% less memory with validation accuracy ~2 percentage points below baseline.

DOMINO | 2nd Place | ILS 4,000 Team Prize

GitHub

- Developed a Flutter emergency-response application with Firebase as its database backend, Hebrew speech transcription, LLM parsing of casualty reports, synced field/command/hospital dashboards, and medical handoff PDFs.

Computer Architecture Simulation Suite

GitHub

- Built L1/L2 cache and two-level branch-predictor simulators (LRU, write-back/write-allocate, L-share/G-share), plus an out-of-order RAW dependency analyzer. Evaluated miss rates, access times, and latency-aware critical paths.

TECHNICAL SKILLS

Languages & systems: C/C++, Python, x86 assembly, Bash, Linux, computer architecture, firmware analysis.

Performance & parallel computing: Performance profiling, bottleneck analysis, benchmarking, parallel programming, CUDA kernel optimization, OpenMP, OpenCL.

ML frameworks & libraries: PyTorch, Hugging Face Transformers, NumPy.

Tools & infrastructure: Docker, Kubernetes, Git, GitHub Actions (CI/CD), Make, CMake.

Deep learning optimization: Familiar with KV/prefix caching, PagedAttention, continuous batching, chunked prefill, speculative decoding, and data/tensor/pipeline parallelism.

EDUCATION & HONORS

Technion, Israel Institute of Technology

2022 - Mar 2027 (expected)

B.Sc. in Computer Engineering | Average: 86.6/100 · **Dean's List, Spring 2023/24**

Selected coursework:

High Performance Parallel Computing (99) · Deep Learning on Computational Accelerators (94) · Computational Accelerators and Accelerated Systems (93) · Performance Engineering of Software Systems (91) · Concurrent and Distributed Programming for Data Processing and Machine Learning (90).



Semester Excellence

THIS IS TO CERTIFY THAT

Tal Ben Ami

ID: 212525257

Mr. Tal Ben Ami was included in the Dean's list of honors for scholastic achievement in the Spring semester of 2023/24.

Dr. Efrat Nativ-Ronen



Academic Secretary Undergraduate Studies

Haifa, 25.09.2026



Transcript of **TAL BEN AMI** ID: **212525257**

who studies

for the degree **BACHELOR OF SCIENCE IN COMPUTER ENGINEERING**

in the Andrew and Erna Viterbi Faculty of Electrical and Computer Engineering

and accumulated **142.5** credit points out of **158.5** credit points required for a degree, with a GPA of **86.6**

SUBJECT	CREDITS	GRADE	SEMESTER
00440102 Safety in Ee Labs.	0	Pass	2022-2023 Winter
00440105 Theory of Electronic Circuits	4	90	2023-2024 Winter
00440127 Basics of Semiconductor Devices	3.5	96	2023-2024 Spring
00440131 Signals and Systems	5	83	2023-2024 Spring
00440157 Electrical Engineering Lab 1a	2	96	2024-2025 Summer
00440167 Lab. Project a	4	97	2025-2026 Winter
00440252 Digital Systems and Computer Structure	5	80	2022-2023 Spring
00450002 An Overview of Electrical and Comp.Eng.	1	99	2022-2023 Winter
00460267 Computer Architecture	3	90	2024-2025 Winter
00460278 Computational Accelerators and Accelera	3	93	2025-2026 Spring
00460279 High Performance Parallel Computing	3	99	2025-2026 Spring
01030015 Introductory Mathematics	0	Exemption without points	2022-2023 Winter
01040013 Differential and Integral Calculus 2T	5.5	77	2022-2023 Spring
01040016 Algebra 1/Extended	5	85	2022-2023 Winter
01040034 Introduction to Probability H	3.5	81	2022-2023 Summer
01040036 DIFFERENTIAL AND INTEGRAL CALCULUS 1T	5	63	2022-2023 Winter
01040038 ALGEBRA 2M	2.5	Pass	2022-2023 Spring
01040134 Modern Algebra H	2.5	79	2024-2025 Winter
01040136 ORDINARY DIFFERENTIAL EQUATIONS M	4	Pass	2022-2023 Spring
01040214 Fourier Series and Integral Transforms	2.5	84	2023-2024 Winter
01040215 Complex Functions a	2.5	78	2023-2024 Winter
01040220 Partial Differential Equations/T	2.5	90	2023-2024 Winter
01130013 Introductory Physics 1	0	Exemption without points	2022-2023 Winter

(E): The course was taught in English

Minimal Passing Grade is 55

Grade Scale: 0 – 100

Haifa, 25.09.2026

Page 1 of 3



Transcript of **TAL BEN AMI** ID: **212525257**

SUBJECT	CREDITS	GRADE	SEMESTER
01130014 Introductory Physics 2	0	Exemption without points	2022-2023 Winter
01140071 Physics 1M	3.5	88	2022-2023 Winter
01140073 Quantum physics for engineering	3.5	68	2023-2024 Spring
01140075 Physics 2Mm	5	86	2022-2023 Spring
01230015 Introductory Chemistry	0	Exemption without points	2023-2024 Winter
02340117 Introduction to Computer Science H	4	88	2022-2023 Winter
02340118 Computer Organization and Programming	3	100	2023-2024 Spring
02340123 Operating Systems	4.5	93	2024-2025 Winter
02340124 Introduction to Systems Programming	4	96	2023-2024 Winter
02340141 Combinatorics for Cs	3	65	2023-2024 Winter
02340218 Data Structures 1	3	89	2023-2024 Spring
02340247 Algorithms 1	3	87	2024-2025 Winter
02340901 Workshop in Competitive Programming	3	90	2024-2025 Spring
02360013 SELECTED TOPICS IN PERFORMANCE ENGINEERING OF SOFTWARE SYSTEMS	2	91	2025-2026 Spring
02360334 Introduction to Computer Networks	3	94	2024-2025 Spring
02360343 Theory of Computation	3	80	2024-2025 Spring
02360370 Concurrent and Distributed Programming for Data Processing and Machine Learning	3	90	2025-2026 Winter
02360496 Reverse Engineering	3	94	2024-2025 Spring
02360501 Introduction to Artificial Intelligence	3	92	2024-2025 Spring
02360700 Software Design	3	73	2025-2026 Winter
02360781 DEEP LEARNING ON COMPUTATION (E)	3	94	2025-2026 Winter
03200101 Community Program 1	1	Exemption with points	2022-2023 Winter
03200102 Community Program 2	1	Exemption with points	2022-2023 Winter
03240032 Advanced English - a	0	Exemption without points	2023-2024 Winter
03240033 Technical English-Advanced B	3	Exemption with points	2022-2023 Winter
03240053 Hebrew 3	0	Exemption without points	2023-2024 Winter

(E): The course was taught in English

Minimal Passing Grade is 55

Grade Scale: 0 – 100

Haifa, 25.09.2026

Page 2 of 3



Transcript of **TAL BEN AMI** ID: **212525257**

SUBJECT	CREDITS	GRADE	SEMESTER
03240540 Legal Aspects of Busine.Entrepreneurship	2	96	2025-2026 Spring
03240879 Issues in Contemporary Israeli Society (E)	2	97	2025-2026 Winter
03250000 SEXUAL HARASSMENT PREVENTION	0	Pass	2023-2024 Winter
03940805 Physical Education - Athletics	1	98	2025-2026 Winter
03940808 Physical Education-for New Students	1	95	2022-2023 Winter

END OF TRANSCRIPT

I hereby confirm that the information above is correct

Dr. Efrat Nativ Ronen,
Academic Secretary, Undergraduate Studies

